

Source Potapov factorisation

the Euler product of unipotent mass matrices is the Cauchy m -function of X , not the Pick kernel of $\mathfrak{T}_0$

Stefan Hamann

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Abstract

The reviewer architecture (DCCXCVII) asks for a source-pure rational transfer $\Theta_k = \Theta_\infty \star \prod_{p \leq 2^{2k}} \Theta_{p,k}$ whose finite Pick matrix on the canonical nodes is congruent to $I - \mathfrak{T}_0^* \mathfrak{T}_0$ (the r409 graph resolvent). Each prime-power block is a finite J -factor built only from $(p, r, \log p, \Lambda = \log p, \text{Fold, Tent, Arch, Border})$; the constructor may not read eigenvalues of C , the SVD of $\mathfrak{T}_0$, target Schur complements, or Z_{loc} . The unipotent Redheffer product of point-mass 2×2 factors *is* the Cauchy m -function of the X -measure (exact over $\mathbb{Q}$), and Lagrange S_0 at interpolating depth $d_0 = |Y|$ is the Cauchy–Pick kernel of the $\pi_Y(X)^2$ -dressed X -measure (exact). That Pick kernel is positive definite, while $I - \mathfrak{T}_0^* \mathfrak{T}_0$ has one negative square on every one-defect window: Gate 3 fails. The one negative square is the interpolant residual $D_0 = W_Y - S_0$, not an Euler factor. No RH claim.

Status. Lemmas 1–2 are proved (finite identities). Propositions 1–2 are finite machine checks. Ausgang SOURCE_FACTOR_NOT_FOUND. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The factor

Let X be the occupied dual nodes of a window, Y the holes, $u^\vee$ the r356 dual weights, and $d_0 = N_w - 3$ the interpolating depth of $\mathfrak{T}_0$ (r409). A source atom $n = p^r$ contributes a unipotent 2×2

$$E(z; x, w) = \begin{pmatrix} 1 & 0 \\ w/(x - z) & 1 \end{pmatrix}, \quad (1)$$

with x the folded position of $r \log p$ and w the corresponding dual (or Λ) weight. The prime block is the Redheffer product (ordinary 2×2 multiplication)

$$\Theta_{p,k}(z) = \prod_{r: p^r \leq 2^{2k}} E(z; x_{p^r}, w_{p^r}), \quad (2)$$

and Θ_∞ is the same factor for the Arch / completion atom. The matrices E commute: $E(z; x, w) E(z; x', w') = E(z; x', w') E(z; x, w)$, and

$$\Theta_\infty \star \prod_p \Theta_{p,k} = \begin{pmatrix} 1 & 0 \\ m_X(z) & 1 \end{pmatrix}, \quad m_X(z) = \sum_{a \in X} \frac{w_a}{x_a - z}. \quad (3)$$

The finite Pick matrix of this Θ on the Y -nodes is the Cauchy kernel

$$(P_X)_{ij} = \sum_{a \in X} \frac{w_a}{(x_a - y_i)(x_a - y_j)}. \quad (4)$$

2 Exact identities over $\mathbb{Q}$

The r409 five-atom toy, labelled by primes 2, 2², 3, 5 and an Arch atom, is the $\mathbb{Q}$ -test.

Lemma 1 (Redheffer = Cauchy). *On the toy, the product (3) equals m_X exactly. At $z = 7/3$ both sides equal $-7471/22950$. The factors of $p = 2$ and ∞ (the X -atoms) already exhaust the product; $p = 3, 5$ sit on Y .*

Lemma 2 (Cauchy–pi formula for Lagrange S_0). *At interpolating depth $d_0 = |Y|$ (unique Lagrange interpolant, kernel dimension 0),*

$$S_0 = \Delta^{-1} C \Pi W_X \Pi C^T \Delta^{-1},$$

where $C_{ia} = 1/(x_a - y_i)$, $\Pi = \text{diag}(\pi_Y(x_a))$, $\pi_Y(x) = \prod_k (x - y_k)$, and $\Delta = \text{diag}(\pi'_Y(y_j))$. Equivalently S_0 is the Cauchy–Pick kernel of the $\pi_Y(X)^2$ -dressed X -measure, conjugated by Δ . The identity is exact over $\mathbb{Q}$. The graph defect is the residual $D_0 = W_Y - S_0$, congruent to $I - \mathfrak{T}_0^* \mathfrak{T}_0$.

3 Gate 3

Proposition 1 (Pick of the product is not the graph kernel). *On the $d_0 = |Y|$ slice of the toy, D_0 has inertia $(1, 1, 0)$ (the $(2, 2)$ entry equals $-1744/2025$), while P_X has inertia $(2, 0, 0)$. The two kernels are not congruent. On flagship MAIN ($k_z = 9$): $\|\mathfrak{T}_0\| = 1.08014$, $\text{ind}_-(I - \mathfrak{T}_0^* \mathfrak{T}_0) = 1$, while P_X is positive definite ($\lambda_{\min}(P_X) = 8.677$). The same mismatch holds at $k_z = 15$ ($\|\mathfrak{T}_0\| = 1.10672$) and $k_z = 18$ ($\|\mathfrak{T}_0\| = 1.24966$).*

Remark 1. A positive X -measure produces a Herglotz m_X and a positive Pick kernel, independently of how the masses are grouped by primes. The one negative square of GraphResolvent is the interpolant residual $D_0 = W_Y - S_0$ in $P_{<d_0}$, which mixes all primes through the polynomial space. No local Euler factor sees it.

4 Index anatomy

Proposition 2 (the one square is not a prime prefix). *Growing- Y Uvarov innovations of $\mathfrak{T}_0$, taken in source order (holes attributed to the nearest prime power, then p ascending, r within), have zero negative pivots on w_9 (minimum $3.25 \cdot 10^{-7}$). The r409 position-order chain has one negative pivot at index 36: the sign of a sequential Uvarov pivot is order-dependent and does not localise the defect to a prime block. Lag-local interpolants $\text{fold}(c_p^P)$ (the tent deposits of a single prime, same X/Y mask) have $\text{ind}_- = 18, 19, 15$ for $p = 2, 3, 5$, versus $\text{ind}_- = 1$ for the full operator — the source product is not a simpler index structure (stop rule 3 of DCCXCVII).*

Permute of Λ -weights on the fixed mask sends $\text{ind}_-(I - \mathfrak{T}^* \mathfrak{T})$ from 1 to 20; Gaussian jitter to 19; scramble to 21. The Euler product remains of index 0 (positive masses). The r289 rational twin keeps $\text{ind}_- = 1$ and $\|\mathfrak{T}_0\|$ bit-near 1.08014.

5 What is a theorem, what is closed

The source Potapov *product* as a factorisation of m_X is a theorem (Lemmas 1–2). As a factorisation of GraphResolvent it is closed: Gate 3 fails on every one-defect window tested, the constructor is source-pure (AST: no $\text{eig}(C)$, no SVD of $\mathfrak{T}_0$, no Z_{loc}), and the candidate index is simpler than the operator only because it is the wrong object. R432 (Krein–Langer index one of Θ) is blocked by the contract (requires FACTOR.EXACT). P1 stays $\lambda_2(C) \geq 1$ on the dual quotient. No RH claim.